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Experiment 10: 5G Network Slicing Environment Using MATLAB

Aim : To analyze the performance of a 5G Network Slicing environment by studying the number of network slices, slice utilization, and end-to-end latency using MATLAB.

Objectives :

Objective – 1: Analyze the Number of Network Slices Analyze the creation and allocation of multiple network slices in a 5G network using MATLAB. Observe how different slices can be configured to support diverse service requirements and applications.

Objective – 2: Compare Slice Utilization (%) Compare the utilization of different network slices using MATLAB. Analyze how efficiently network resources are allocated and utilized across multiple slices.

Objective – 3: Analyze End-to-End Latency (ms) Analyze the end-to-end latency of different network slices using MATLAB. Compare the latency values to understand the Quality of Service (QoS) requirements of various 5G applications.

Procedure:

Objective 1

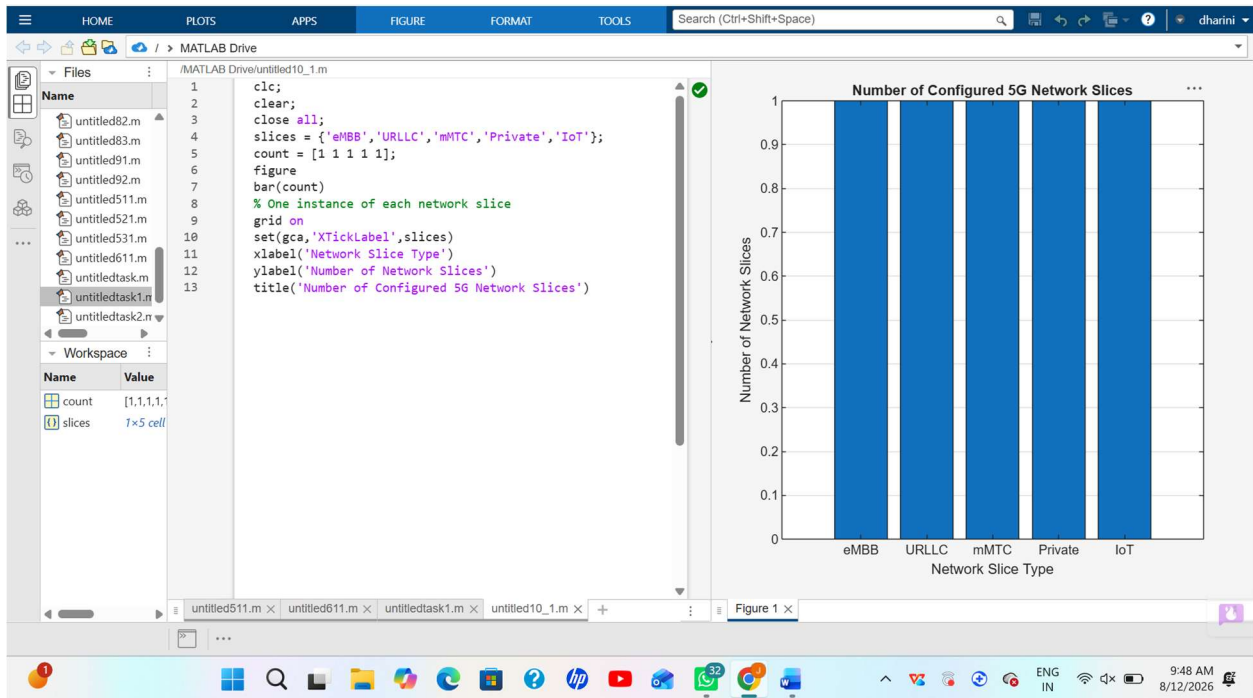
```
clc;
clear;
close all;
slices = {'eMBB','URLLC','mMTC','Private','IoT'};
count = [1 1 1 1 1];
figure
bar(count)
% One instance of each network slice
grid on
set(gca,'XTickLabel',slices)
```

```
xlabel('Network Slice Type')
```

```
ylabel('Number of Network Slices')
```

```
title('Number of Configured 5G Network Slices')
```

Output:



Objective 2:

```
clc;
```

```
clear;
```

```
close all;
```

```
slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});
```

```
utilization = [85 70 60 75 50];
```

```
figure;
```

```
bar(slices, utilization);
```

```
grid on;
```

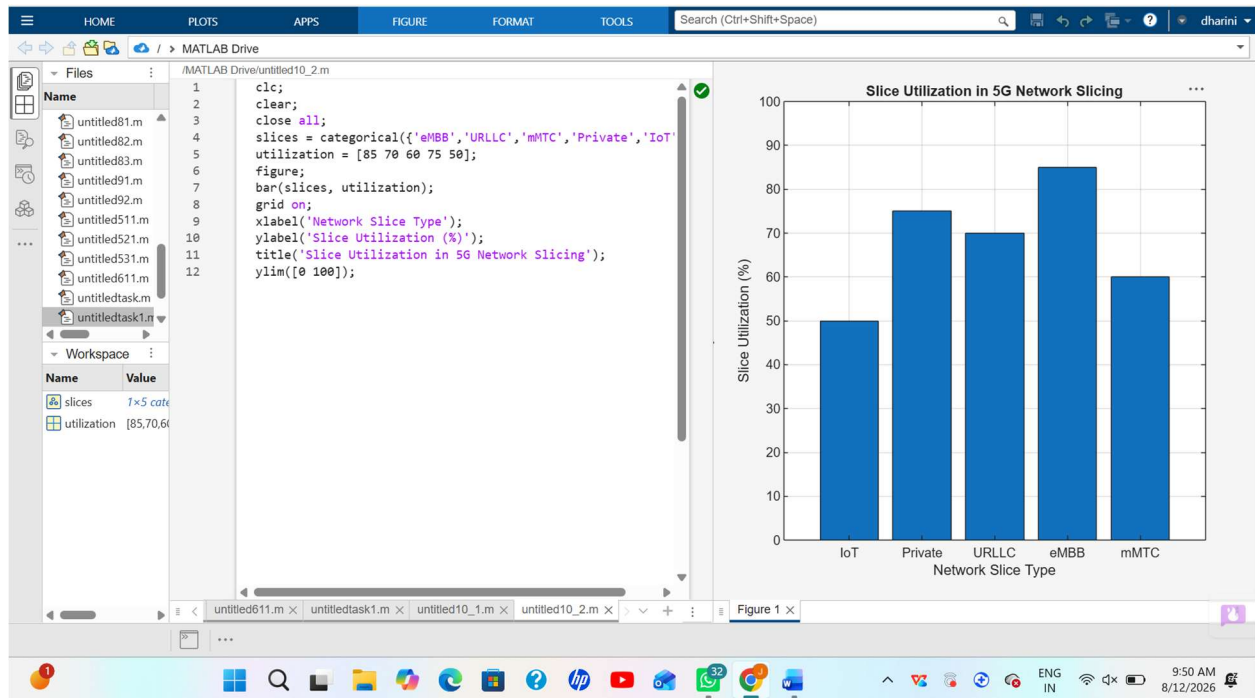
```
xlabel('Network Slice Type');
```

```
ylabel('Slice Utilization (%)');
```

```
title('Slice Utilization in 5G Network Slicing');
```

```
ylim([0 100]);
```

Output:



Objective 3:

```
clc;
```

```
clear;
```

```
close all;
```

```
slices = [1 2 3 4 5];
```

```
latency = [25 5 40 15 60];
```

```
figure;
```

```
plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);
```

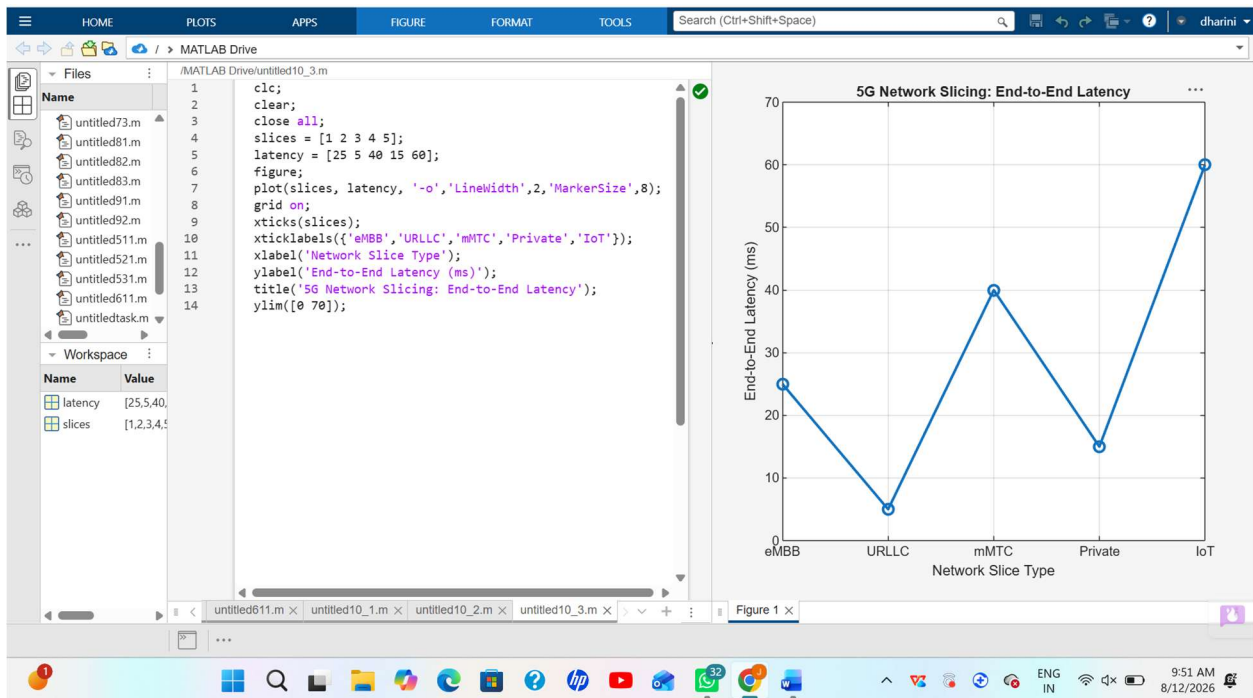
```
grid on;
```

```
xticks(slices);
```

```
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
```

```
xlabel('Network Slice Type');  
ylabel('End-to-End Latency (ms)');  
title('5G Network Slicing: End-to-End Latency');  
ylim([0 70]);
```

Output:



Result: The MATLAB analysis successfully demonstrated the characteristics of a 5G Network Slicing environment. The results showed that multiple network slices can coexist while exhibiting different utilization levels and latency characteristics based on their intended services.